Question ID: 2c288148

Assessment	Test	Domain	Skill	Difficulty
SAT	Math	Advanced Math	Nonlinear equations in one variable and systems of equations in two variables	Hard

Question

$$\sqrt{k-x}=58-x$$

In the given equation, k is a constant. The equation has exactly one real solution. What is the minimum possible value of $4k$?

Correct Answer: 231

Rationale

The correct answer is 231. It's given that $\sqrt{k-x}=58-x$. Squaring both sides of this equation yields $k-x=(58-x)^2$, which is equivalent to the given equation if $58-x>0$. It follows that if a solution to the equation $k-x=(58-x)^2$ satisfies $58-x>0$, then it's also a solution to the given equation; if not, it's extraneous. The equation $k-x=(58-x)^2$ can be rewritten as $k-x=3{,}364-116x+x^2$. Adding x to both sides of this equation yields $k=x^2-115x+3{,}364$. Subtracting k from both sides of this equation yields $0=x^2-115x+(3{,}364-k)$. The number of solutions to a quadratic equation in the form $0=ax^2+bx+c$, where a, b, and c are constants, can be determined by the value of the discriminant, b^2-4ac. Substituting -115 for b, 1 for a, and $3{,}364-k$ for c in b^2-4ac yields $(-115)^2-4(1)(3{,}364-k)$, or $4k-231$. The equation $0=x^2-115x+(3{,}364-k)$ has exactly one real solution if the discriminant is equal to zero, or $4k-231=0$. Subtracting 231 from both sides of this equation yields $4k=231$. Dividing both sides of this equation by 4 yields $k=57.75$. Therefore, if $k=57.75$, then the equation $0=x^2-115x+(3{,}364-k)$ has exactly one real solution. Substituting 57.75 for k in this equation yields $0=x^2-115x+(3{,}364-57.75)$, or $0=x^2-115x+3{,}306.25$, which is equivalent to $0=(x-57.5)^2$. Taking the square root of both sides of this equation yields $0=x-57.5$. Adding 57.5 to both sides of this equation yields $57.5=x$. To check whether this solution satisfies $58-x>0$, the solution, 57.5, can be substituted for x in $58-x>0$, which yields $58-57.5>0$, or $0.5>0$. Since 0.5 is greater than 0, it follows that if $k=57.75$, or $4k=231$, then the given equation has exactly one real solution. If $4k<231$, then the discriminant, $4k-231$, is negative and the given equation has no solutions. Therefore, the minimum possible value of $4k$ is 231.